Class-Imbalance Mitigation with Frozen Readouts and Selective Low-Rank Adaptation

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Abstract

Earlier mitigation studies used low-rank adaptation (LoRA) to adapt the Virchow2 encoder for every method, including the cross-entropy reference. These training procedures yielded a weaker reference under balanced classes and less measured damage from class imbalance. This study evaluates the mitigations in their intended form: methods without a representation-learning stage use frozen features; only classifier re-training (cRT), Gaussian clouded logit adjustment (GCL) and its two-stage variant use LoRA. The main reference is the damage studies' frozen logistic regression, which loses 7.65 percentage points of balanced accuracy on BRACS and 2.91 on TCGA-UT as the class ratio increases from 1 to 100. The curriculum of data augmentation uses a separate, matched frozen-encoder reference trained by stochastic gradient descent. Balanced sampling and logit adjustment on frozen features recover the most: 4.42 and 3.75 points on BRACS, and 2.39 and 2.16 points on TCGA-UT. The LoRA methods achieve similar imbalance-specific recovery, but their deficit of about 3.5 points under balanced classes offsets most of that recovery on BRACS. This deficit combines representation and readout differences and cannot be attributed to LoRA alone. No method fully recovers the damage.

1 Motivation

The damage studies measured the loss in patient-macro balanced accuracy when a linear readout of frozen Virchow2 features (Zimmermann et al., 2024) trains on imbalanced classes (Queisler, 2026e,d). At a class ratio of $\rho = 100$, the loss is 7.65 points on BRACS and 2.91 points on TCGA-UT. The mitigation studies then tested eight mitigations, using low-rank adaptation (LoRA) (Hu et al., 2022) for every method, including the cross-entropy (CE) reference (Queisler, 2026b,c). LoRA was intended only for methods with a representation-learning stage. This training procedure changed the reference: under balanced classes on BRACS, LoRA CE reached 36.53 points compared with the frozen readout's 39.57, reducing the measured damage. A frozen-encoder follow-up showed that some methods maintain or improve accuracy without LoRA (Queisler, 2026a).

This study asks: *how much of the measured damage does each mitigation recover when applied as intended?* The damage studies' frozen logistic regression provides the main reference; the curriculum of data augmentation (CUDA) uses a matched frozen-encoder reference, as described below.

2 Design

The study follows the damage studies' design: Virchow2, three patient-disjoint splits with ten patient draws each (30 shards per dataset), and a fixed patch budget allocated at balanced ($\rho = 1$) and severe ($\rho = 100$) class ratios. Validation and test patients, class order, and class-to-rank permutations remain the same. Table 1 summarizes the method implementations. Classifier

re-training (cRT), Gaussian clouded logit adjustment (GCL) and its two-stage variant (GCL2) use LoRA.

Method	Representation	Readout	Strength grid at $\rho = 100$
CE (reference)	frozen	logistic regression	none
Logit adjustment (LA)	frozen	CE logits $-\tau \log \hat{\pi}$	$\tau \in \{0.5, 1, 1.5, 2\}$; TCGA-UT adds 0.25
Balanced sampling (BS)	frozen	class-weighted logistic regression	$s \in \{0.5, 1\}$
DisAlign	frozen	learned calibration of the CE head	$\gamma \in \{0.5, 1, 1.5, 2\}$
cRT	LoRA (CE)	re-trained linear head, balanced sampling	none
GCL	LoRA (GCL loss)	its own cosine head	σ grid of the mitigation studies
GCL2	LoRA (GCL loss)	re-trained cosine head, GCL loss	as GCL
CUDA	frozen	linear head, augmented images	$p_{\text{aug}} \in \{0.25, 0.5, 1\}$

Table 1: Methods without a representation-learning stage use frozen Virchow2 features. Classifier re-training (cRT), Gaussian clouded logit adjustment (GCL) and its two-stage variant (GCL2) (Kang et al., 2020; Li et al., 2022) retain the low-rank adaptation (LoRA) first stage from the mitigation studies. CE: cross-entropy; LA: logit adjustment; BS: balanced sampling; CUDA: curriculum of data augmentation. $\hat{\pi}$ denotes realized training class shares.

Frozen readouts. The CE reference uses the damage studies' multinomial logistic regression. Validation selects the L2 regularization strength λ from nine values between 10^{-6} and 100. Logit adjustment (LA) (Menon et al., 2021) subtracts $\tau \log \hat{\pi}$ from the CE logits and re-selects λ for each τ. Adding the offset to the logits during training gives the same predictions: with an unpenalized intercept, the offset changes only the optimal intercept. Train-time and post-hoc adjustment therefore constitute one method here. Balanced sampling (BS) uses class weights $w_c \propto n_c^{-s}$, which yield the same expected loss as sampling. DisAlign (Zhang et al., 2021) trains its calibration module on the frozen features and the CE head at the selected λ, following the mitigation studies' second-stage procedure.

LoRA methods. cRT, GCL and GCL2 reuse the mitigation studies' results unchanged at both class ratios. These methods train their heads by stochastic gradient descent (SGD), whereas the frozen-feature methods above use logistic regression. Their comparison therefore includes differences in both representation and readout training. The balanced gain G defined below measures the combined difference under balanced classes; subtracting it does not isolate the effect of LoRA.

CUDA. The curriculum of data augmentation (CUDA) (Ahn et al., 2023) requires encoder forward passes on images at every training step, which logistic regression on fixed features cannot provide. CUDA therefore retains the follow-up study's frozen-encoder SGD training procedure (Queisler, 2026a). Its reference at both class ratios is a CE head trained by the same procedure without augmentation (CE_{SGD}). The $\rho = 100$ CUDA results come from the follow-up study.

Reference reproduction. The reproduction criterion requires each CE refit to select the same λ as the damage studies' stored fit and reproduce its test balanced accuracy within 0.01

Table 2: Cross-entropy (CE) references: patient-macro balanced accuracy (BA, %) for logistic regression (LR) and stochastic gradient descent (SGD). r1 and r100 denote class ratios $\rho = 1$ and $\rho = 100$. Damage $D = \mathrm{BA(r1)} - \mathrm{BA(r100)}$, with 95 % intervals; * material (point estimate ≥ 1 percentage point, lower bound > 0).

Dataset	LR r1	LR r100	D	SGD r1	SGD r100	D_{SGD}
BRACS	39.57	31.92	7.65 [5.72, 9.81]*	38.52	31.19	7.34 [5.29, 9.53]*
TCGA-UT	72.06	69.14	2.91 [2.51, 3.33]*	71.51	67.31	4.20 [3.77, 4.62]*

points. On each dataset, 57 of the 60 shard-ratio combinations match every prediction. Five others (two on BRACS, three on TCGA-UT) retain λ but differ on 1 to 68 test patches, changing balanced accuracy by at most 0.004 points. One BRACS combination retains λ but differs on 38 of 22,133 test patches, changing balanced accuracy by 0.056 points; the analysis retains it as an exception. These differences arise because the damage studies' solver stops before the optimum, making predictions near the decision boundary sensitive to floating-point order.

3 Evaluation

The evaluation separates recovery under severe imbalance from gains that also occur under balanced classes. Using patient-macro balanced accuracy (BA) in percentage points, pooled over the 30 shards, the damage D, recovery R of method m, recovery share R/D, balanced gain G and imbalance-specific recovery $R - G$ are

$$D = \mathrm{BA}_{\mathrm{CE}}(\rho{=}1) - \mathrm{BA}_{\mathrm{CE}}(\rho{=}100), \qquad R = \mathrm{BA}_m(\rho{=}100) - \mathrm{BA}_{\mathrm{CE}}(\rho{=}100), \qquad R/D, \tag{1}$$

$$G = \mathrm{BA}_m(\rho{=}1) - \mathrm{BA}_{\mathrm{CE}}(\rho{=}1), \qquad R - G. \tag{2}$$

G measures the method–reference difference under balanced classes; $R - G$ measures how that difference changes under severe imbalance. Subtracting G removes the difference observed at balance, but any readout effect that varies with the class ratio can remain in $R - G$. For LA and BS, $G = 0$ by construction: equal class counts make the offset constant and the weights equal. For CUDA, $\mathrm{CE}_{\mathrm{SGD}}$ replaces CE throughout these contrasts. Validation BA alone selects each method's strength per shard.

Uncertainty estimation follows the mitigation studies: Bayesian bootstrap weights on test patients (Rubin, 1981) combine with resampling of draws within each split over 10,000 replicates. Shared weights pair every contrast. Each contrast has a 95% percentile interval without multiplicity correction, since the study makes no joint claim. A recovery R (or $R-G$) is *material* if its point estimate is at least one point and its lower bound exceeds zero. A method recovers the damage *fully* if the interval for R/D contains one. The study made no advance prediction about the size of the recovery.

4 Results

The frozen reference reproduces the damage studies (Table 2). The SGD reference loses a similar amount on BRACS (7.34 points), but more on TCGA-UT (4.20 versus 2.91 points), where its $\rho = 100$ head is 1.83 points below logistic regression. The $\mathrm{CE}_{\mathrm{SGD}}$ references trained on a central processing unit (CPU) in the follow-up study and on a graphics processing unit (GPU) here differ by at most 0.02 points of BA.

Frozen methods. Balanced sampling and logit adjustment recover the most on both datasets (Table 3): BS recovers 4.42 points on BRACS and 2.39 on TCGA-UT; LA recovers 3.75 and

Table 3: Recovery R, recovery share R/D, balanced gain G and imbalance-specific recovery $R-G$, with 95 % intervals. Accuracy contrasts are in percentage points; R/D is dimensionless. LA: logit adjustment; BS: balanced sampling; cRT: classifier re-training; GCL: Gaussian clouded logit adjustment; GCL2: its two-stage variant; CUDA: curriculum of data augmentation. CUDA uses a cross-entropy reference trained by stochastic gradient descent ($\mathrm{CE}_{\mathrm{SGD}}$). * material (point estimate ≥ 1 percentage point, lower bound > 0).

Dataset	Method	R	R/D	G	$R-G$
BRACS	LA	3.75 [2.17, 5.58]*	0.49 [0.33, 0.64]	0 (by construction)	3.75 [2.17, 5.58]*
BRACS	BS	4.42 [2.90, 6.12]*	0.58 [0.43, 0.72]	0 (by construction)	4.42 [2.90, 6.12]*
BRACS	DisAlign	0.65 [0.32, 1.00]	0.09 [0.04, 0.13]	-0.06 [-0.12, -0.01]	0.72 [0.38, 1.07]
BRACS	cRT	0.54 [-0.61, 1.69]	0.07 [-0.09, 0.20]	-3.52 [-4.92, -2.21]	4.05 [2.39, 5.82]*
BRACS	GCL	1.14 [-0.20, 2.57]	0.15 [-0.03, 0.32]	-3.37 [-4.73, -2.04]	4.51 [2.81, 6.32]*
BRACS	GCL2	1.61 [-0.11, 3.56]	0.21 [-0.02, 0.40]	-3.68 [-5.06, -2.34]	5.29 [3.11, 7.83]*
BRACS	CUDA	1.82 [0.81, 2.95]*	0.25 [0.12, 0.39]	-0.39 [-0.82, 0.06]	2.21 [1.15, 3.41]*
TCGA-UT	LA	2.16 [1.87, 2.47]*	0.74 [0.68, 0.81]	0 (by construction)	2.16 [1.87, 2.47]*
TCGA-UT	BS	2.39 [2.04, 2.75]*	0.82 [0.77, 0.87]	0 (by construction)	2.39 [2.04, 2.75]*
TCGA-UT	DisAlign	1.00 [0.87, 1.14]*	0.34 [0.29, 0.40]	0.01 [-0.00, 0.02]	0.99 [0.86, 1.13]
TCGA-UT	cRT	1.49 [0.99, 2.01]*	0.51 [0.36, 0.67]	-0.96 [-1.41, -0.50]	2.46 [1.97, 2.96]*
TCGA-UT	GCL	1.68 [1.22, 2.17]*	0.58 [0.44, 0.71]	-1.10 [-1.48, -0.71]	2.78 [2.23, 3.34]*
TCGA-UT	GCL2	1.95 [1.39, 2.50]*	0.67 [0.51, 0.81]	-1.15 [-1.54, -0.74]	3.09 [2.50, 3.68]*
TCGA-UT	CUDA	0.89 [0.49, 1.33]	0.21 [0.12, 0.30]	-0.03 [-0.15, 0.09]	0.92 [0.54, 1.32]

2.16 points. These gains represent 82% and 74% of the damage on TCGA-UT, and 58% and 49% on BRACS, respectively. Both methods mainly improve recall in the rarest third of classes, defined by training patch counts. On BRACS, recall rises from 7.3% under CE to 25.5% under LA and 19.1% under BS; on TCGA-UT, it rises from 60.1% to 68.4% and 68.2%, respectively. DisAlign recovers little: 0.65 points on BRACS and 1.00 point on TCGA-UT. The latter just meets the materiality rule, whereas its imbalance-specific recovery (0.99) does not.

LoRA methods. At $\rho = 100$, cRT, GCL and GCL2 do not materially outperform the frozen reference on BRACS (0.54 to 1.61 points, all intervals containing zero), but do on TCGA-UT (1.49 to 1.95 points). Their balanced-class deficits help explain the smaller recovery on BRACS. At $\rho = 1$, the LoRA methods fall 3.4 to 3.7 points below the frozen reference on BRACS and 1.0 to 1.2 points below it on TCGA-UT. Their imbalance-specific recovery $R-G$ ranges from 4.05 to 5.29 points on BRACS and from 2.46 to 3.09 points on TCGA-UT. GCL2 has the largest point estimate on both datasets. On BRACS, GCL also exceeds BS (4.51 versus 4.42 points), whereas cRT falls below it (4.05 points); on TCGA-UT, all three LoRA methods exceed BS (2.39 points). GCL2 also achieves the highest recall in the rarest third of classes (27.3% on BRACS, 72.2% on TCGA-UT). On BRACS, the deficit under balanced classes offsets most of the improvement in relative performance under imbalance. This deficit reflects the combined training procedure, including representation and readout differences, rather than an isolated cost of LoRA.

CUDA. Against its SGD reference, CUDA recovers 1.82 points on BRACS (25% of D_{SGD}, material) and 0.89 points on TCGA-UT (21%, not material). Its balanced gain is close to zero. At $\rho = 1$, validation selects the weakest augmentation ($p_{\mathrm{aug}} = 0.25$) in 29 of 30 shards on each dataset.

Full recovery and selection. No method fully recovers the damage. BS achieves the highest recovery share on both datasets: 0.58 [0.43, 0.72] on BRACS and 0.82 [0.77, 0.87] on TCGA-UT. Validation selects the maximum BS strength, $s = 1$, in 28 of 30 BRACS shards and all TCGA-UT shards; this is the natural endpoint of its range. Two methods select an open grid

boundary, as in the mitigation studies: DisAlign selects the largest $\gamma = 2$ in 13 of 30 BRACS shards, and GCL2 selects the smallest $\sigma = 0.01$ in 16 of 30 TCGA-UT shards. The grids were not extended.

Choice of reference. For methods evaluated against frozen logistic regression, replacing that reference with the mitigation studies' LoRA CE reference shifts each recovery by the same amount: 0.34 points higher on BRACS, where LoRA CE is weaker at $\rho = 100$, and 0.81 to 0.82 points lower on TCGA-UT, where it is stronger (e.g. LA recovers 1.35 instead of 2.16 points). The ranking among these methods remains unchanged. CUDA retains its matched SGD reference.

5 Limitations

The study reuses data already inspected in the damage and mitigation studies and is therefore not an independent test. It covers one encoder and one severe class ratio. The frozen and LoRA methods differ in both readout (logistic regression versus SGD heads) and representation. Comparing their recoveries therefore combines both differences; $R - G$ removes only the component visible under balanced classes and can retain readout effects that depend on the class ratio. CUDA uses a different reference from the other methods. The design does not compare frozen and LoRA-adapted encoders within cRT or GCL, so it cannot attribute the balanced-class deficit or the imbalance-specific recovery to LoRA alone.

The analysis uses two acceptance rules relaxed after their first evaluation. The initial check of train-time and post-hoc logit adjustment compared a fit at the damage studies' solver tolerance with a tightly converged fit and failed on both datasets. At matched precision, the two agree on every test patch. Reference reproduction initially required identical test predictions; the 0.01-point tolerance and the single reported exception replaced that rule. Both changes have the same cause: the damage studies' logistic-regression solver stops early. Refitting the reference at tight tolerance would change BRACS reference accuracy by about half a point on one examined shard; the resulting change in D is unknown.

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